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Predictability Is Not Substitutability

A cost-of-substitution framework for H&E-based molecular prediction across 5 cancers

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Long abstract
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Summary. Discrimination (AUROC) shows that a molecular marker is *predictable* from H&E; it does not show that the prediction can *substitute* for the molecular test. Under one protocol, 10 sealed-forward endpoints in lung, gastric and head-and-neck cancer resolved as **7 candidates (1 confirmed, 6 undecided)** and **3 positive controls**. A near-ceiling AUROC (0.939, lung histology) was excluded by a site audit; the six unconfirmed candidates were undecided rather than negative.

1. Background

H&E models predict MSI, driver mutations and subtypes at high reported AUROC [1–3], but AUROC does not say which treatment a patient receives when the prediction is wrong [4].

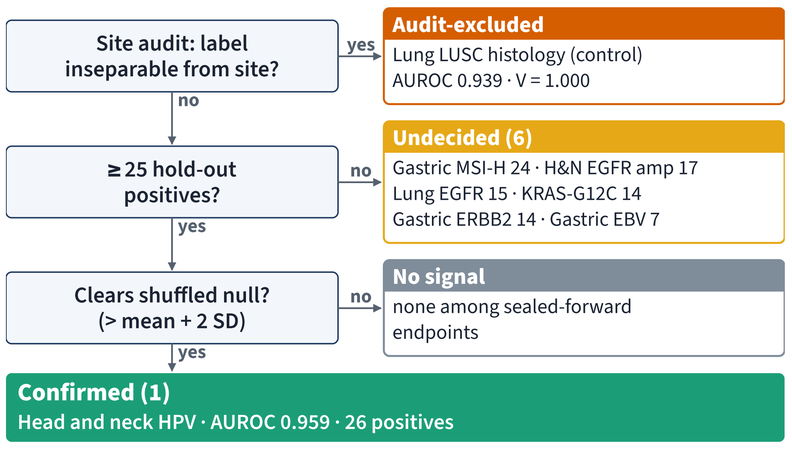
In **our prior work**, an H&E recurrence model reached sensitivity 0.857 for low-risk but 0.529 for high-risk patients [5]. Such predictors are often confounded [6]; in a 33-task benchmark, mean AUROC fell from 0.689 to 0.630 externally [7].

2. Cost of substitution

Under a routing fixed in advance, the cost for reference axis a is the mean treatment distance d (0–1) between reference and routed axis over its n_a slides:

$$cost_a = (1/n_a)\ \Sigma_{i:\ a_i=a}\ d(a,\ \hat{a}_i)$$

The **primary measure** is the distance-free **misassignment rate**. It prices an error at the moment of assignment, not drug response.



Controls: H&N grade passed; gastric Lauren failed (95% CI includes 0.5); lung histology above.

Verdict rules. Counts are the 10 sealed-forward endpoints of Figure 2. Gastric ERBB2 is classified Undecided.

3. Methods

- **Cohorts:** breast (TCGA-BRCA anchor, ~1,010 slides), lung, colorectal, gastric, head and neck.
- **Pre-processing:** 256 × 256 tiles at 20×, Otsu segmentation, ≤ 5,000 tiles/slide.
- **Model:** UNI v1 embeddings (1,024-d) [10] + CLAM-SB attention MIL [11].
- **Evaluation:** site-disjoint hold-out only.
- **Controls:** 5-seed label-shuffled null (AUROC > null mean + 2 SD); prevalence baseline; histology-only / pixel-mean baseline (reported, not a verdict criterion); 1,000× bootstrap CI.
- Predictions **sealed by commit** before hold-out results were produced (lung, gastric, head and neck); the V = 1 audit-exclusion rule was formalized after the site audit.
- **Scope:** 10 sealed-forward endpoints (incl. 3 positive controls); breast/colorectal (4) and endometrial (1) reported but not counted.

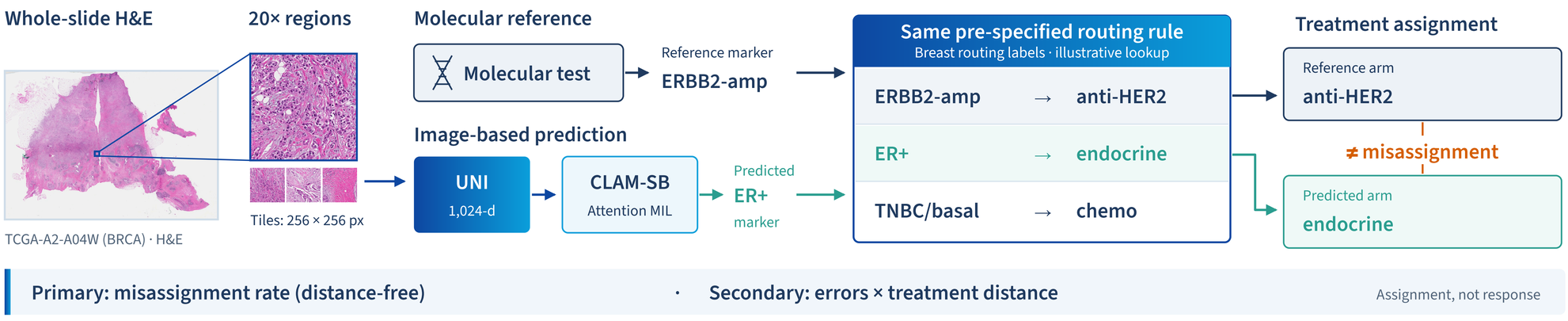


Figure 1. Framework. The molecular result and the H&E prediction pass through the same pre-specified treatment routing; a disagreement in the assigned arm is a misassignment. Breast-anchor misassignment: see Figure 5.

4. Results

Table 1. Selected sealed-forward hold-out endpoints: lung, gastric, head and neck.

Endpoint	AUROC	95% CI	Hold-out pos.	Verdict
Head and neck HPV	0.959	0.921–0.986	26	Confirmed‡
Lung LUSC histology (control)	0.939	0.905–0.967	153	Audit-excluded
Gastric MSI-H	0.860	0.759–0.941	24	Undecided
Lung EGFR activating	0.852	0.725–0.953	15	Undecided
Lung KRAS-G12C	0.681	0.577–0.783	14	Undecided
Gastric ERBB2 amplification	0.644	0.523–0.771	14	Undecided

Also undecided (Figure 2): H&N EGFR amp (17 pos.), gastric EBV (7). Not counted (anchor/retrospective): breast HER2 0.599; colorectal anti-EGFR 0.705 (84 pos.), MSI-high 0.918 (21), BRAF V600E 0.882 (15). ‡ Primary model (UNI). With 20 label shuffles HPV cleared the null under UNI, UNI2-h and Virchow2 [12] (5 shuffles: Virchow2 0.9199 vs null threshold 0.9234). HPV pixel-mean baseline AUROC 0.922 [0.854–0.975].

Confirmation. Head and neck HPV was the only non-control endpoint meeting the primary rule. It has known histological correlates (non-keratinising, basaloid morphology) [8] and HPV association has been predicted from H&E in two cohorts of 273 patients [8].

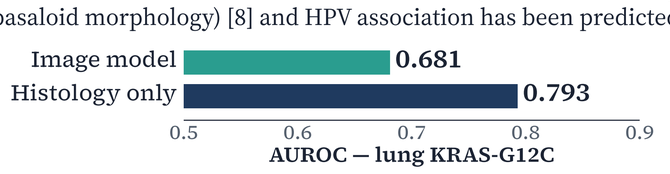


Figure 4. Baseline check: histological subtype alone, with no image, beats the image model for lung KRAS-G12C. Remains Undecided (14 positives): this baseline is a reported control, not a verdict rule, so the contrast is an observation only.

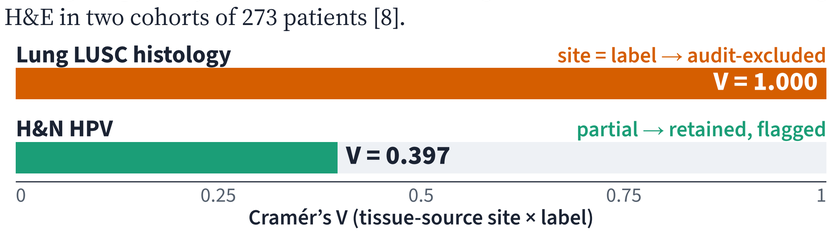


Figure 3. Site-confounding audit. TCGA institution codes coincide exactly with LUAD/LUSC status [9], so the 0.939 AUROC was audit-excluded; HPV shows a partial association.

4a. Positive-count ceiling

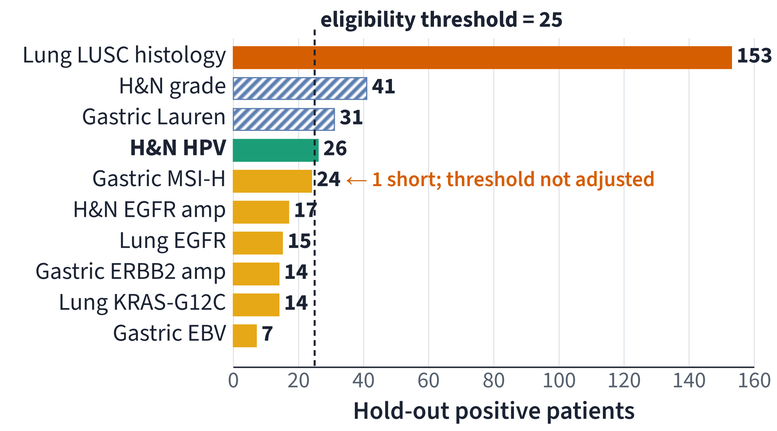


Figure 2. Hold-out positives vs the eligibility threshold of 25 (hatched: positive controls; lung histology, also a control, shown as audit-excluded). All six non-HPV candidate endpoints fall below it — undecided, not negative. Gastric MSI-H fell one short; the criterion was not adjusted.

4b. Breast anchor

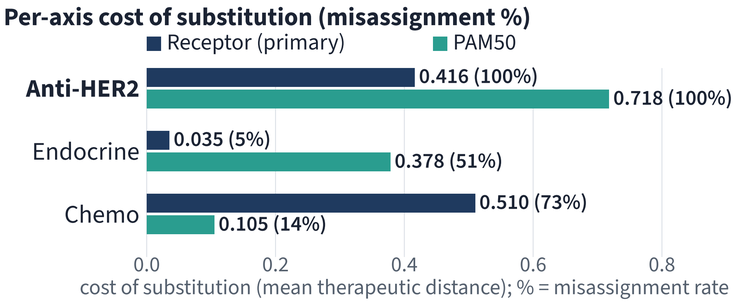


Figure 5. Per-axis substitution cost under two pre-specified routings. In external CPTAC-BRCA, every anti-HER2 slide is misassigned under both; endocrine/chemotherapy costs reverse between routings (no safety claim). The anti-HER2 − endocrine contrast excludes 0 under both (receptor 0.381 [0.331–0.427]; PAM50 0.340 [0.251–0.425]). In an external pCR cohort (exploratory), the anti-HER2 prediction did not stratify response (AUROC 0.533).

5. Discussion

Retrospective and hypothesis-generating. Site-disjoint splits block leakage, not confounding. Positive count alone does not explain the pattern (Figure 6).

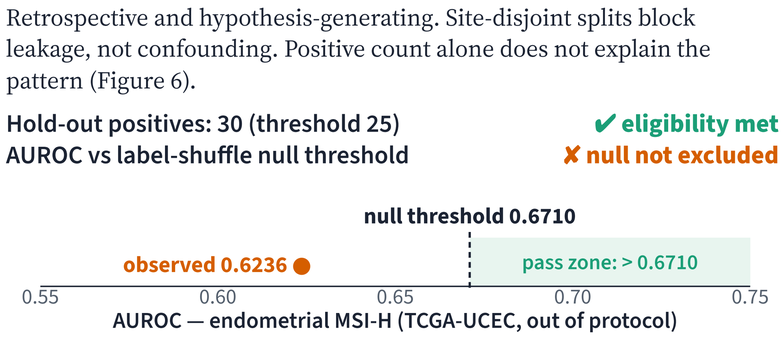


Figure 6. Eligibility alone is not enough. Out-of-protocol endometrial MSI-H met the 25-positive criterion yet failed null exclusion; excluded from the confirmation tally.

6. Conclusion

1. Explicit verdict rules separated **1 confirmed** and **6 undecided** candidates, and audit-excluded a near-ceiling control.
2. The framework rejected a near-ceiling AUROC and, under the primary model, retained one literature-supported endpoint.
3. Reporting the undecidable as **undecidable**, not negative, is the central point; the procedure transfers to any proposed H&E surrogate.

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Code & results
github.com/Geongyu/predictability-substitutability